



Prince Saiyad <prince.saiyad@renataiot.com>

Front end review_Accumulator Server

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22 December 2025 at 10:44

To: prince.saiyad@renataiot.com

Cc: anil.nehra@marelli.com, simar.kaur@external.marelli.com, rajendra.kumar@renataiot.com

Greetings!

Dear Prince,

Just briefing about OEE calculations as we discussed in meeting.

% OEE = % (AVAILABILITY x PERFORMANCE x QUALITY)

Availability= Operating time/Loading time

- Operating time = Loading time - (breakdown time/stop time/AM activities)
- Loading time = 430 min that is fixed

For Example, we have stopped machine for 20 min. then Operating time = 430 - 20 i.e 410 min.

Availability = Operating time/Loading time => 410/430 = 95.35 %, but in this case, challenge is that how to calculate the down time automatically.

We can follow below steps as per my understanding :

- Calculate cycle time & accordingly no of parts that should be produced.
- how many parts are left per shift to produce could be converted back to minutes and that time could be considered as breakdown time.

Performance= Net Operating time/ Operating time

- From previous example considering Operating time = 410 min.
- Net Operating time = Operating Time - Performance loss

Challenge is to feed the performance loss, so in that case we can consider “(breakdown time/stop time/AM activities)” as a performance loss.

- Net Operating time => 430 - 20 = 410 min , considering 20 min down time as performance loss, same as in availability example.

Performance = 410/410 = 100%

Quality= Good Unit/Total unit produced

For example, Total unit produced = 100, out of them 02 are NG/Pending then good parts => 100 - 2 = 98 nos.

Quality = 98/100 = 98%

OOE= 95.35%x100%x98%

OOE =93.44%

Piston Pre Assy		50	0.83
Piston Oring Lubrication		42	0.70
Snap Ring Preassy		30	0.50
Final Assy + Gas Filling		140	2.33
HP Leak Test		70	1.17
Functional Test (EOL)		60	1.00
Micro -Percussion Marking		31	0.52
External O-ring Assy		35	0.58
External Oring Test (Leakage Test)		46	0.77
Check List & Packaging			0.00
Painting		39	0.65
Server			0.00
			0.00
Turning Piston		130	2.17
Turning Housing		255	4.25
Honing machine 1#		135	2.25
Honing machine 2#		73	1.22
Deburring		32	0.53
Piston marking		40	0.67
Final Washing			0.00

Regarding CT (Min) we can consider time between first scan and second scan of same part no.

Productivity= Output/Input {on the basis of total parts OK/(OK+NG)}

Regards.

Nirman

Regards.

Nirman

-----Original Appointment-----

From: Kumar Nirman

Sent: Friday, December 19, 2025 10:52 PM

To: prince.saiyad@renataiot.com; Nehra Anil; Kaur Simar (Ext)

Subject: FW: Front end review_Accumulator Server

When: Friday, December 19, 2025 11:30 AM-12:00 PM (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi.

Where: Microsoft Teams Meeting

Importance: High

[Quoted text hidden]